Experiment 2

Title

Docker Installation, Configuration, and Running Images

Objective

To pull Docker images, run containers with port mapping, and manage the container lifecycle using Docker commands.

Tools Used

- Docker
- Docker Desktop
- macOS Terminal
- Nginx Docker Image

Theory

Docker is a containerization platform that allows applications to run in isolated environments called containers. Docker images act as templates for containers. Containers can be started, stopped, and removed easily using Docker lifecycle commands.

Procedure

Step 1: Pull Docker Image

The Nginx Docker image was pulled from Docker Hub.

```
~/desktop — -zsh                                  ~ — -zsh

Last login: Sat Feb  7 00:55:09 on ttys000
pratiksmac@MacBook-3 ~ % docker --version

Docker version 29.2.0, build 0b9d198
pratiksmac@MacBook-3 ~ % docker pull nginx

Using default tag: latest
latest: Pulling from library/nginx
5380ffdd215e: Pull complete
0e9ba714bdf1: Pull complete
58072fb4fd7c: Pull complete
1063666c8b8e: Pull complete
3ea009573b47: Pull complete
fb8853dab2da: Pull complete
efc7865cf13e: Pull complete
ee66459a851f: Download complete
875ddf4f8f5c: Download complete
Digest: sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
pratiksmac@MacBook-3 ~ %
```

Step 2: Run Docker Container with Port Mapping

A Docker container was started using the Nginx image with port mapping.

```
~/desktop — -zsh                                  ~ — -zsh

Last login: Sat Feb  7 00:55:09 on ttys000
pratiksmac@MacBook-3 ~ % docker --version

Docker version 29.2.0, build 0b9d198
pratiksmac@MacBook-3 ~ % docker pull nginx

Using default tag: latest
latest: Pulling from library/nginx
5380ffdd215e: Pull complete
0e9ba714bdf1: Pull complete
58072fb4fd7c: Pull complete
1063666c8b8e: Pull complete
3ea009573b47: Pull complete
fb8853dab2da: Pull complete
efc7865cf13e: Pull complete
ee66459a851f: Download complete
875ddf4f8f5c: Download complete
Digest: sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
pratiksmac@MacBook-3 ~ % docker run -d -p 8080:80 nginx

cb44fcf207b7e9cf346b9b19d84185618ad380f2f88db28a59004e34a12338e2
pratiksmac@MacBook-3 ~ %
```

Step 3: Verify Running Containers

The list of running containers was checked.

```
Last login: Sat Feb  7 00:55:09 on ttys000
pratiksmac@MacBook-3 ~ % docker --version

Docker version 29.2.0, build 0b9d198
pratiksmac@MacBook-3 ~ % docker pull nginx

Using default tag: latest
latest: Pulling from library/nginx
5380ffdd215e: Pull complete
0e9ba714bdf1: Pull complete
58072fb4fd7c: Pull complete
1063666c8b8e: Pull complete
3ea009573b47: Pull complete
fb8853dab2da: Pull complete
efc7865cf13e: Pull complete
ee66459a851f: Download complete
875ddf4f8f5c: Download complete
Digest: sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
pratiksmac@MacBook-3 ~ % docker run -d -p 8080:80 nginx

cb44fcf207b7e9cf346b9b19d84185618ad380f2f88db28a59004e34a12338e2
pratiksmac@MacBook-3 ~ % docker ps

CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                     NAMES
cb44fcf207b7   nginx     "/docker-entrypoint.…"   4 minutes ago   Up 4 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   magical_nobel
pratiksmac@MacBook-3 ~ %
```

Step 4: Stop , Remove Docker Container and Remove Docker Image

The running container was stopped and removed. The Docker image was removed from the system.

```
pratiksmac@MacBook-3 ~ % docker stop cb44fcf207b7
cb44fcf207b7
pratiksmac@MacBook-3 ~ % docker rm cb44fcf207b7
cb44fcf207b7
pratiksmac@MacBook-3 ~ % docker rmi nginx
Untagged: nginx:latest
Deleted: sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
```

Commands Used

```
docker pull nginx
docker run -d -p 8080:80 nginx
docker ps
docker stop <container_id>
docker rm <container_id>
docker rmi nginx
```

Result

Docker images were successfully pulled, containers were executed with port mapping, and container lifecycle commands were performed successfully.